

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing
and Big Data
Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

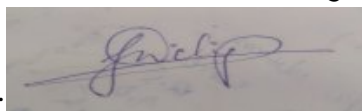
Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **Jani Dilip (192424314)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently.

Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.
Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application

Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. Media Company Video Streaming Platform Migration

Cloud Architecture Design

A scalable cloud architecture for video streaming should include:

Global Content Delivery Network (CDN): Caches video content closer to users, reducing latency.

Load Balancer: Distributes incoming user requests across multiple application servers.

Auto-Scaling Groups: Automatically increase or decrease servers based on demand.

Object Storage: Stores video files reliably (e.g., cloud storage buckets).

Database: Managed cloud database for user accounts and streaming metadata.

Monitoring Service: Tracks performance, bandwidth, and server health.

Role of Load Balancing and Auto-Scaling

Load Balancing

Evenly distributes user requests across available servers.

Prevents server overload.

Improves response time and availability.

Auto-Scaling

Automatically launches new servers during peak traffic.

Removes unused servers during low demand.

Ensures uninterrupted streaming and minimal buffering.

Cost Optimization Strategies

Use CDN caching to reduce bandwidth costs.

Store rarely accessed videos in low-cost archival storage.

Use Auto-Scaling to avoid paying for idle servers.

Purchase Reserved Instances for predictable workloads.

Compress videos using adaptive bitrate streaming to reduce data transfer.

2. Fintech Startup Using Containers Across Multiple Clouds

Proposed Solution

Package applications using Docker containers.

Use Kubernetes for container orchestration across cloud providers.

Deploy clusters in multiple clouds (AWS, Azure, GCP).

Store container images in a centralized container registry.

Use Infrastructure as Code (IaC) for consistent deployment.

Service Discovery and Scaling

Service Discovery

Kubernetes DNS automatically locates services.

Services communicate using internal DNS names.

Scaling

Horizontal Pod Autoscaler (HPA) increases or decreases pods based on CPU or memory usage.

Cluster Autoscaler adds or removes worker nodes automatically.

Benefits of Multi-Cloud

Avoids vendor lock-in.

High availability.
Better disaster recovery.
Improved geographic coverage.
Ability to choose best cloud services.

Risks

Increased management complexity.
Higher networking costs.
Security policy differences.
Difficult monitoring across clouds.
Data synchronization challenges.

3. Backup and Disaster Recovery for E-Learning Platform

Backup and Disaster Recovery Design

Enable automatic cloud backups.
Store backups in multiple geographic regions.
Enable object versioning to recover deleted files.
Configure cross-region replication.
Use snapshot-based backups for databases.
Automate recovery using disaster recovery plans.

Achieving RTO and RPO

Recovery Time Objective (RTO)

Use standby systems.
Automated failover.
Fast restoration from snapshots.
Infrastructure automation.

Recovery Point Objective (RPO)

Frequent incremental backups.
Continuous database replication.

Object versioning.

Cross-region synchronization.

Business Continuity Steps

Schedule regular automated backups.

Replicate data across multiple regions.

Periodically test disaster recovery procedures.

Maintain backup documentation.

Monitor backup success and failures.

Automate failover mechanisms.

Train staff for disaster recovery procedures.

4. Government Hybrid Cloud Architecture

Hybrid Cloud Architecture

Sensitive applications remain in the on-premises data center.

Public cloud hosts non-sensitive applications.

Connect both environments using:

Use API gateways for secure communication.

Centralized monitoring and logging across environments.

Identity Management

Implement Single Sign-On (SSO).

Use Role-Based Access Control (RBAC).

Enable Multi-Factor Authentication (MFA).

Integrate cloud IAM with on-premises Active Directory.

Encryption

Encrypt data at rest using AES-256.

Encrypt data in transit using TLS/SSL.

Use centralized Key Management Services (KMS).

Rotate encryption keys regularly.

Challenges

- Complex infrastructure management.
- Higher operational costs.
- Network latency between environments.
- Security policy consistency.
- Compliance management.
- Difficult monitoring across hybrid environments.

5. SaaS Provider High Availability and Fault Tolerance

Multi-Region Deployment Strategy

- Deploy applications in multiple cloud regions.
- Use global load balancers to route traffic.
- Maintain active-active or active-passive architecture.
- Replicate databases across regions.
- Store backups in separate geographic locations.

Data Replication and Failover

- Configure synchronous replication for critical databases.
- Use asynchronous replication for disaster recovery.
- Enable automatic failover when the primary region fails.
- Perform regular failover testing.
- Use DNS-based traffic routing for regional failures.

Monitoring and Alerting Strategies

- Monitor CPU, memory, disk, and network utilization.
- Track application response times.
- Configure alerts for server failures.
- Monitor database replication status.

Use centralized logging and dashboards.

Enable automatic incident notifications.

Perform regular health checks and uptime monitoring.